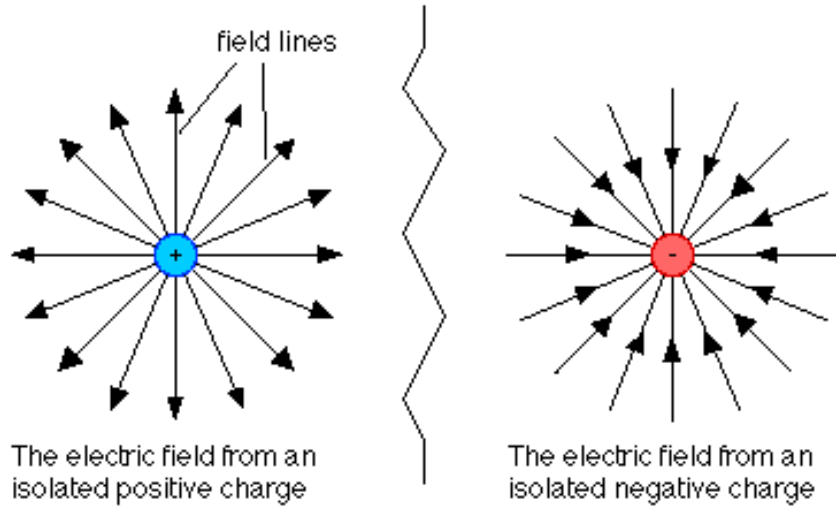


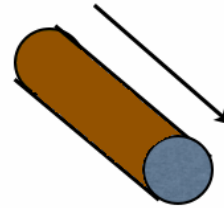
How are electric and magnetic fields linked together?

Stationary Charge → Static Electric Field



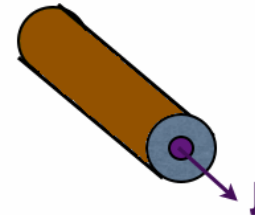
Moving Charges → Current

Current I = Total Flow of Charge Per Time [A]

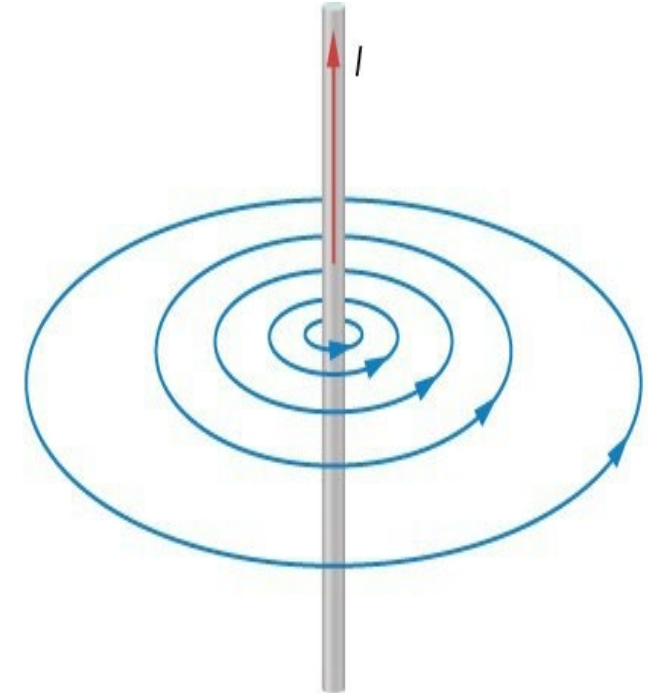


www.maxwells-equations.com

Current Density J = Total Flow of Charge Per Time [A] over a cross section of area [m²]



Current → Magnetic Field





Large-scale Power Generation



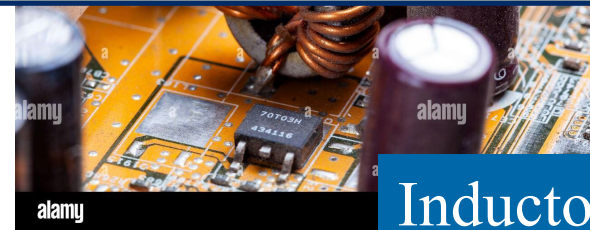
Large-scale manufacturing

Electric Cars

Why may you want to know about the magnetic field?



Transformers



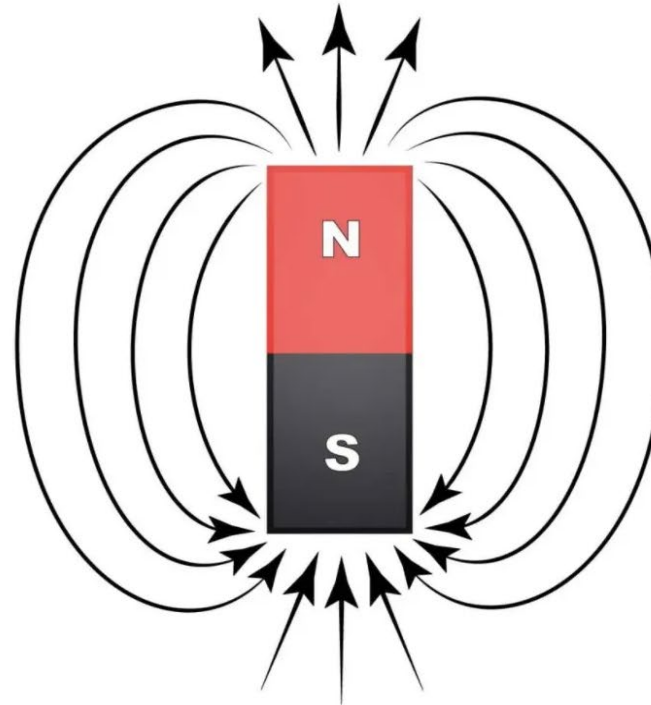
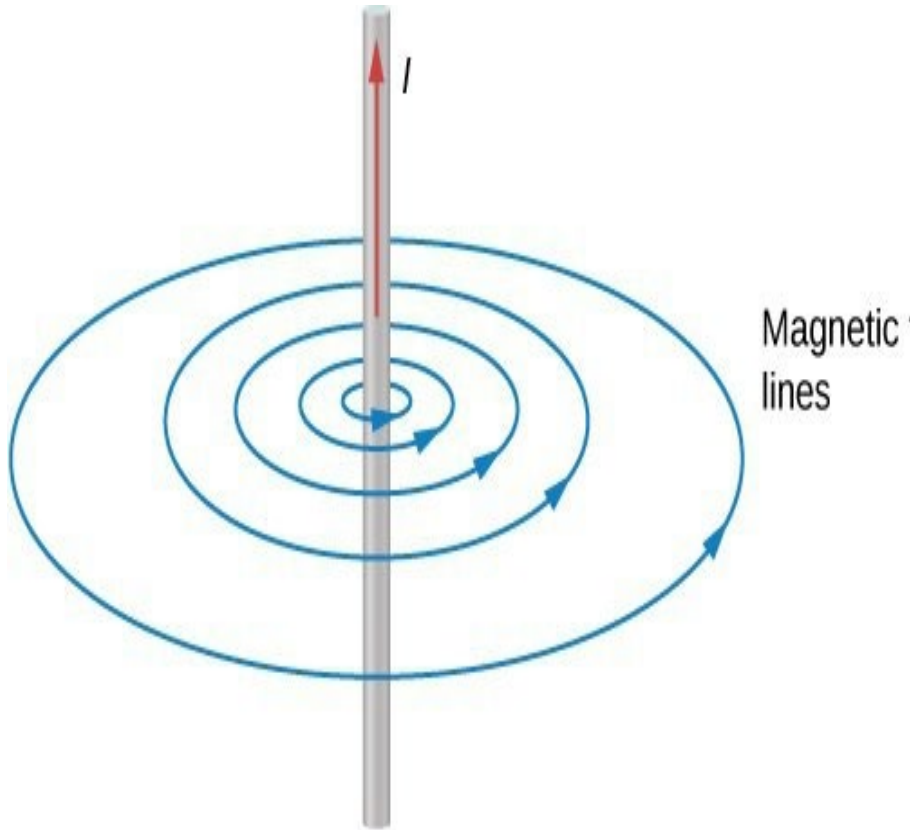
Inductors



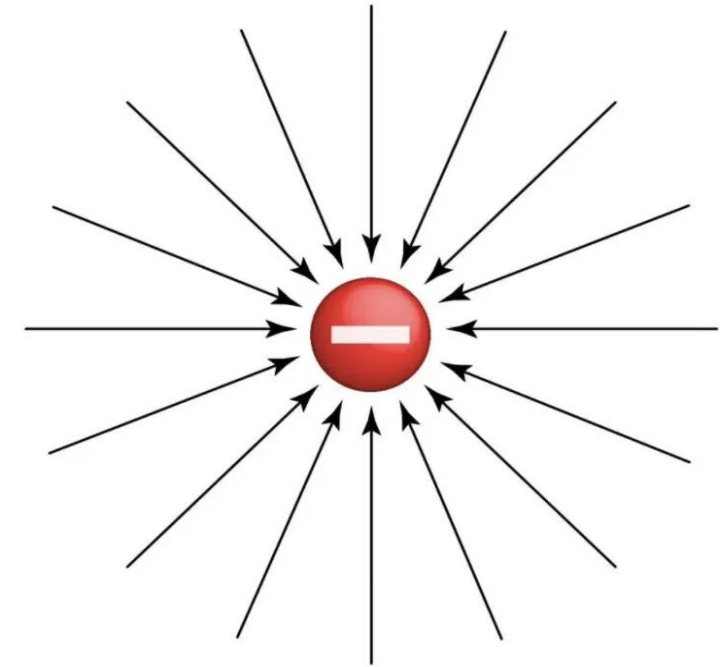
Oersted's experiment (in ~1819)



What is a Magnetic field?



Magnetic field surrounding a magnet



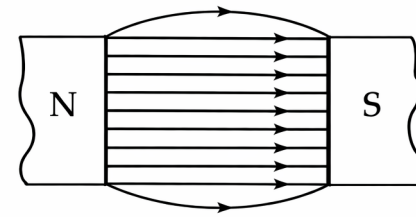
Electric field surrounding a charged particle

It is a force field like the electric field. Visualise the field by the flux lines. A magnetic field is produced by a magnet and by a current.

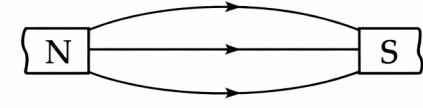
How to measure magnetic field

Two field quantities:

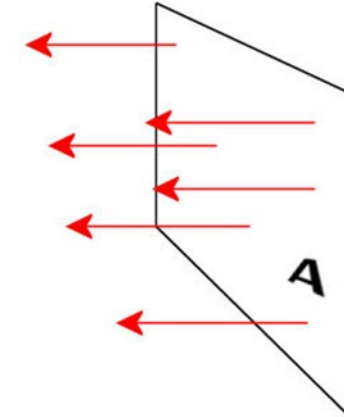
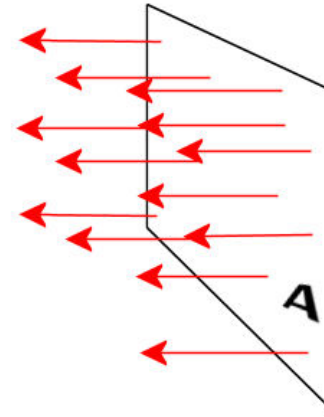
1. Magnetic flux density $\mathbf{B}$
2. Magnetic flux intensity $\mathbf{H}$.



(a) Strong magnetic field



(b) Weak magnetic field



Magnetic field per unit area A is magnetic flux density $\mathbf{B}$, unit is Tesla (T or Wb/m^2).

$\mathbf{H}$, magnetic field intensity measures the strength of the magnetic field. Unit is A/m.

Both $\mathbf{B}$ and $\mathbf{H}$ are vectors and can be specified in a point of the space.

Note the similarity of $\mathbf{B}$ (Wb/m^2) and $\mathbf{H}$ (A/m) with $\mathbf{E}$ (V/m) and $\mathbf{D}$ (C/m^2).

B and **H** are related by the permeability μ (H/m) of the material media.

$$\vec{B} \propto \vec{H} \rightarrow \vec{B} = \mu \vec{H}, \quad \mu = \mu_0 \mu_r$$

where, $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space.

and μ_r is relative permeability of a medium. It is unit less.

In free space or vacuum, $\vec{B} = \mu_0 \vec{H}$

Today's road map

Magnetic field quantities B and H



Postulates (Ampere's law)



B -field calculation



Material response → magnetisation $\mathbf{M}$



H field definition



μ_r and B – H curve

Learning Goals

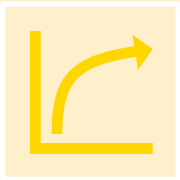
After today's lecture, you will be able to



explain the postulates of magnetostatics, infer their relation to Ampere's law, and **calculate** *magnetic field $\mathbf{B}$* using it.

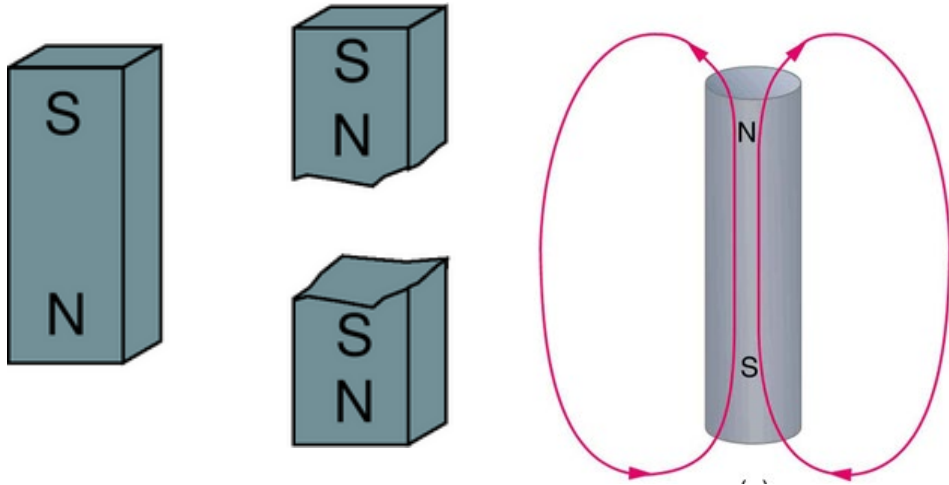


explain and **analyse** the relation between $\mathbf{B}$ and $\mathbf{H}$ using the *concept of permeability* in different material media.



explain the *magnetic saturation and hysteresis* using the concept of magnetization and magnetic dipoles.

Two fundamental postulates of Magnetostatics



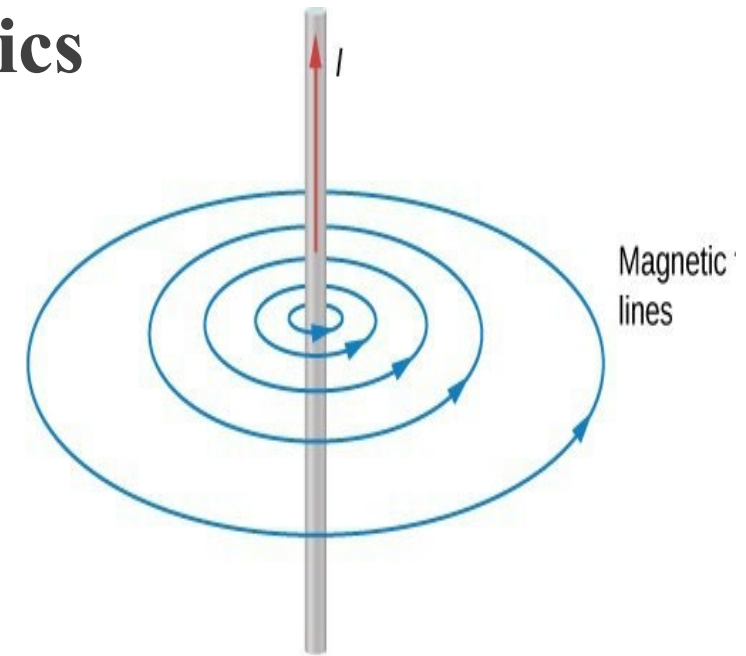
Observation 1: No magnetic monopoles

- Magnetic field lines always form closed loops.
- There is no isolated magnetic pole.
- Total magnetic flux through a closed surface is zero.

$$\vec{\nabla} \cdot \vec{B} = 0$$

EXAM signpost:

You must understand both the physical meaning and mathematical form of these two equations. Everything in this topic flows from them.



Observation 2: Current produces a circulating magnetic field

- A steady current creates a magnetic field
- Field circulates around the current

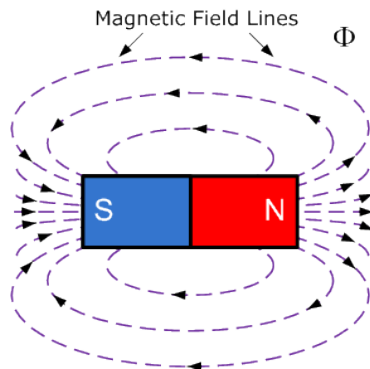
$$\vec{\nabla} \times \vec{B} \neq 0 \rightarrow \vec{\nabla} \times \vec{B} = \mu \vec{J}$$

Compare with Electrostatic

Magnetostatic

$$\vec{\nabla} \bullet \vec{B} = 0$$

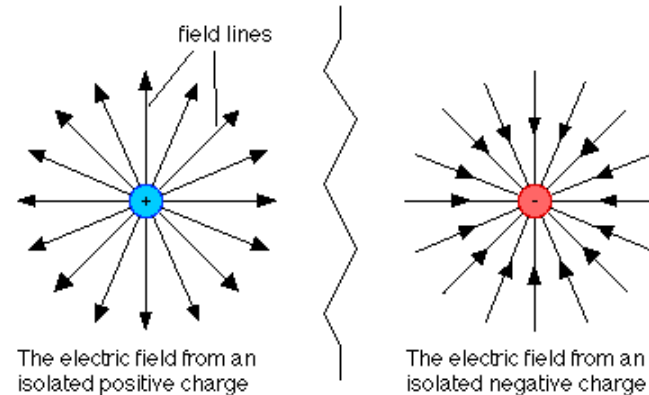
$$\vec{\nabla} \times \vec{B} = \mu \vec{J}$$



Electrostatic

$$\vec{\nabla} \bullet \vec{E} = \frac{\rho}{\epsilon}$$

$$\vec{\nabla} \times \vec{E} = 0$$

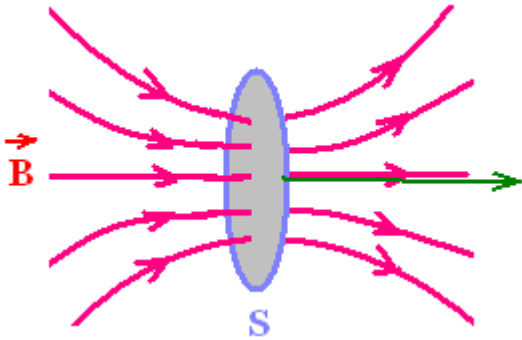


How to use the postulates: integral form and magnetic flux

Applying Divergence theorem to volume integral of the first postulate,

$$\vec{\nabla} \cdot \vec{B} = 0 \rightarrow \int_{vol} \vec{\nabla} \cdot \vec{B} dv = 0 \rightarrow \oint_S \vec{B} \cdot d\vec{s} = 0 \quad (\text{Divergence theorem})$$

Surface integral of flux density $\mathbf{B}$ gives us the total flux passing through a surface area.



$$\text{Magnetic Flux } \Phi = \int_S \vec{B} \cdot d\vec{s} \quad \text{Wb}$$

$$\oint_S \vec{B} \cdot d\vec{s} = 0 \rightarrow \sum \Phi = 0$$

Total flux enclosed in a closed surface is zero.

Flux in a magnetic circuit is analogous to *current* of an electrical circuit.

Integral form of the second postulate and Ampere's law

Applying Stokes' theorem to the surface integral of the second postulate,

$$\vec{\nabla} \times \vec{B} = \mu \vec{J} \rightarrow \int_s (\vec{\nabla} \times \vec{B}) \cdot d\vec{s} = \mu \int_s \vec{J} \cdot d\vec{s} \rightarrow \oint_c \vec{B} \cdot d\vec{l} = \mu I_{enc}$$

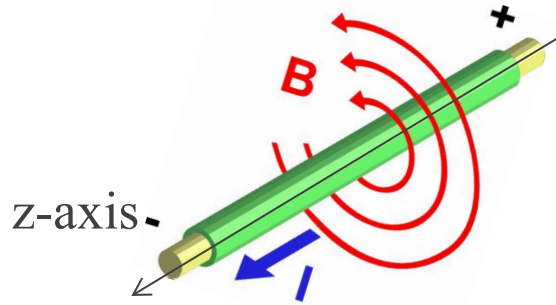

Like Gauss's law in electrostatics, Ampere's law is useful to find the magnetic field.

If we can find an Amperian loop, in which B is constant (or uniform) and in the same direction as the defined loop, the closed line integration mathematically simplifies significantly.

Example 1: Magnetic Field of a long current-carrying conductor

Find the magnetic field inside and outside of a current-carrying conductor using Ampere's law.

Step 1: Identify the coordinate and symmetry

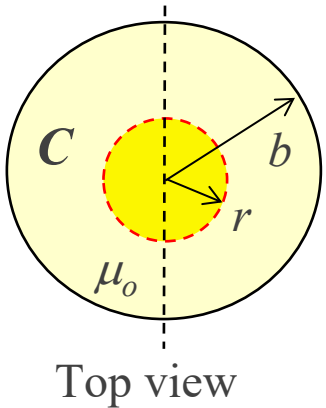


In cylindrical co-ordinates, the conductor of radius b , is along the z direction and carries current I . B is in θ -direction.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu I_{enc}$$

Step 2: Choose an Amperian loop to apply Ampere's law inside the conductor

- A closed contour C of radius r ($0 < r < b$),



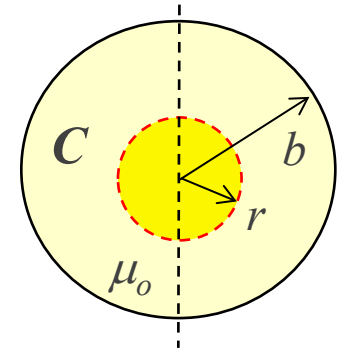
$$\oint_C \vec{B}_{in} \cdot d\vec{l} = \mu I_c \quad \text{where, } I_c \text{ is current enclosed by contour } C.$$

Step 3: Find vector function $\mathbf{B}_{in}$

$$B_{in}(2\pi r) = \mu_0 I_c \rightarrow \therefore B_{in} = \frac{\mu_0 I_c}{2\pi r}, \quad \vec{B}_{in} = \frac{\mu_0 I_c}{2\pi r} \vec{a}_\theta$$

Step 4: Evaluate enclosed current I_c . Is I_c the same as I ?

$$B_{in} = \frac{\mu_0 I_c}{2\pi r}$$



I_c is the current enclosed by the contour C of radius $r < b$.

I is the total current enclosed by the conductor of radius b .

If the current density is uniform for $r < b$

$$I_c = \left(\frac{r^2}{b^2} \right) I$$

Step 5: Substitute I_c in B_{in}

$$B_{in} = \frac{\mu_0 I_c}{2\pi r} = \frac{\mu_0}{2\pi r} \left(\frac{r^2}{b^2} \right) I = \frac{\mu_0 r}{2\pi b^2} I, \quad r \leq b$$

$$\vec{B}_{in} = \frac{\mu_0 r}{2\pi b^2} I \vec{a}_\theta \quad (\mathbf{B}_{in} \text{ is a function of } r \text{ and in the direction of } \theta)$$

Step 6: Find $\mathbf{B}_{\text{out}}$

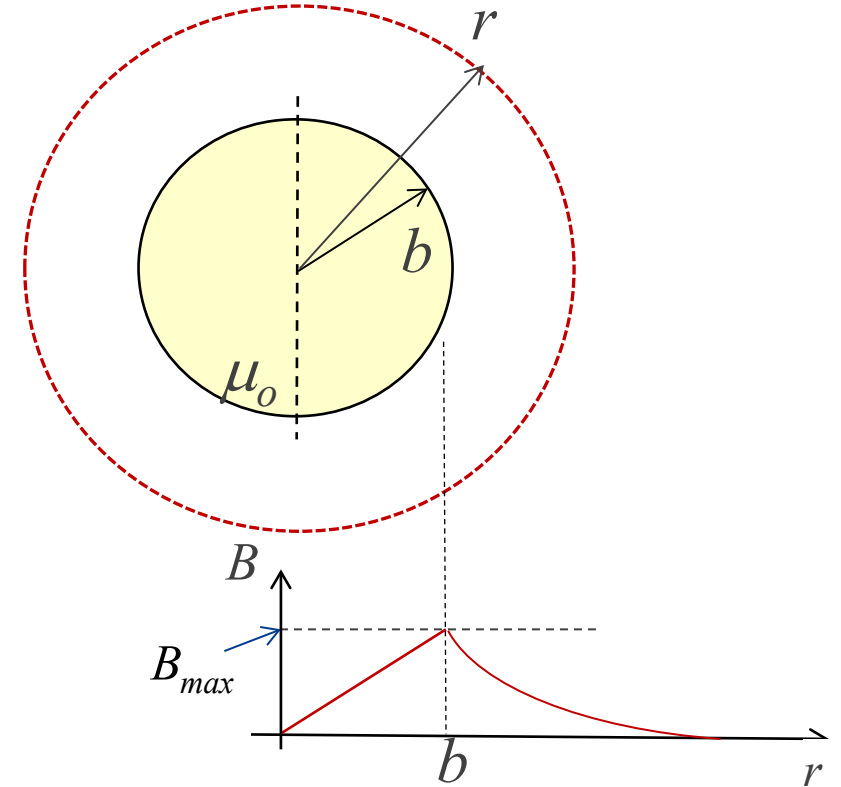
Choose an Amperian loop to apply Ampere's law outside the conductor

- A closed contour C of radius r ($r > b$),

$$\oint_C \vec{B}_{\text{out}} \cdot d\vec{l} = \mu_0 I_{\text{enc}} \rightarrow B(2\pi r) = \mu_0 I_{\text{enc}},$$

$$r > b, \quad I_{\text{enc}} = I, \quad \vec{B}_{\text{out}} = \frac{\mu_0 I}{(2\pi r)} \vec{a}_\theta,$$

($\mathbf{B}_{\text{out}}$ is a function of $1/r$ and in the direction of θ)



Step 7: plots

$$\vec{B}_{\text{in}} = \frac{\mu_0 r}{2\pi b^2} I \vec{a}_\theta, \quad 0 < r < b$$

$$\vec{B}_{\text{out}} = \frac{\mu_0 I}{(2\pi r)} \vec{a}_\theta, \quad r > b$$

$$B_{\text{max}} = \frac{\mu_0 I}{2\pi b}$$

EXAM signpost:

Be able to:

- Identify symmetry
- Choose correct Amperian loop
- Determine enclosed current correctly

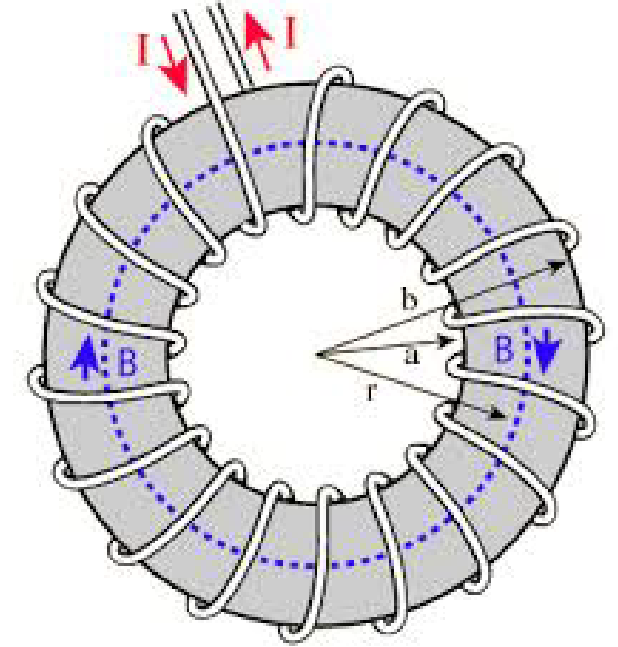
Let's try out (an application example)

Using Ampere's law, show that magnetic field inside an air-cored toroid is given as,

$$B = \frac{\mu_0 NI}{2\pi r} \text{ where } a < r < b$$



● The toroid result is a standard examinable configuration.



N is number of turns in the toroid

Is there any flux density in the inner and outer side of the core?

A contour considered at the inner section of the torus does not include any current, hence the flux density is zero.

A contour considered the outer area includes both current going in and out, and hence, the flux density produced cancels out at this area. The Contour at the mean radius gives us flux density in the core.

Does Ampere's law always work?

$$\oint_c \vec{B} \cdot d\vec{l} = \mu I_{enc}$$

The application of Ampere's law is simple when B is uniform in the contour; otherwise, the closed line integral is more complex. (Non-uniform $B \rightarrow$ line integral difficult)

What is the alternative?

- In electrostatics, we found alternative methods with potential V (e.g., Poisson/Laplace's equations) when a Gaussian surface cannot be found.
- For the magnetic field, to use similar methods, we first need to define a **magnetic potential**.

Magnetic Vector Potential $\vec{A}$

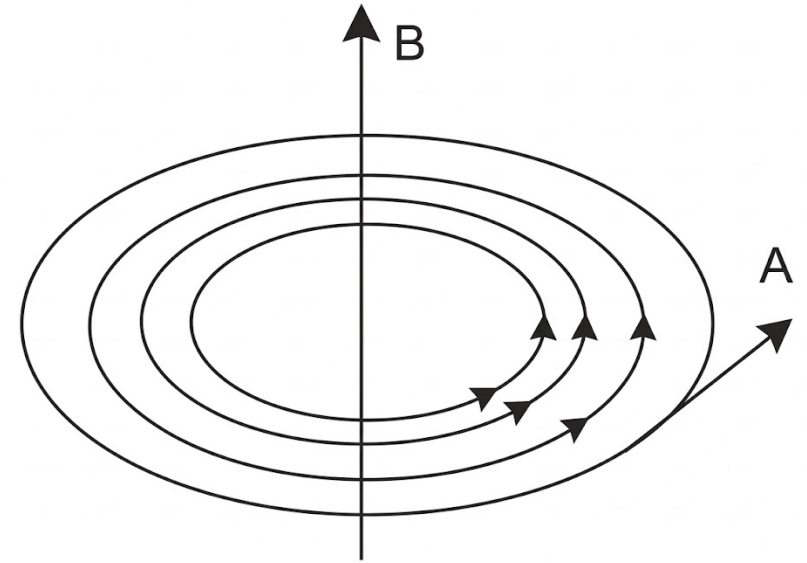
First postulate of magnetostatic, $\vec{\nabla} \bullet \vec{B} = 0$ (1)

The vector null identity, $\vec{\nabla} \bullet \vec{\nabla} \times \vec{A} = 0$ (2)

Comparing (1) and (2), $\vec{B} = \vec{\nabla} \times \vec{A}$

where $\vec{A}$ is called vector magnetic potential. Unit is Wb/m

Note that magnetic potential $\vec{A}$ is a vector whereas electric potential V is a scalar. Unlike V , $\vec{A}$ is not physically measurable quantity.



Poisson's equation for magnetic field

Second postulate: $\vec{\nabla} \times \vec{B} = \mu \vec{J}$ But $\vec{B} = \vec{\nabla} \times \vec{A}$

Sub **B** in second postulate, $\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \mu \vec{J}$

But from vector identity, $\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \vec{\nabla}^2 \vec{A} \quad \therefore \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \vec{\nabla}^2 \vec{A} = \mu \vec{J}$

However, $\vec{\nabla} \cdot \vec{A} = 0$ (for the same reason for which Div. of **B** is zero.)

$$\therefore \vec{\nabla}^2 \vec{A} = -\mu \vec{J}$$

Vector Poisson's equation

$$\vec{\nabla}^2 V = -\frac{\rho}{\epsilon}$$

Scalar Poisson's equation

Solving Poisson's equation, **A** can be found, and **B** can be calculated:

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

Also, flux can be found directly from **A**. Take surface integration of both sides and apply Stokes' theorem to the RHS, $\int_s \vec{B} \cdot d\vec{s} = \int_s (\vec{\nabla} \times \vec{A}) \cdot d\vec{s} \rightarrow \Phi = \oint_c \vec{A} \cdot d\vec{l}$

See examples 6.4, 6.5, 6.6, and 6.7 in your textbook.

Pause & Reflect

(Before we take a short break...)

 **Take 1 minute to jot down:**

One idea you just learned

One step you could explain to a friend

One word that describes how confident you feel right now
(scale: 😞 *unsure* → 😊 *okay* → 😄 *confident*)

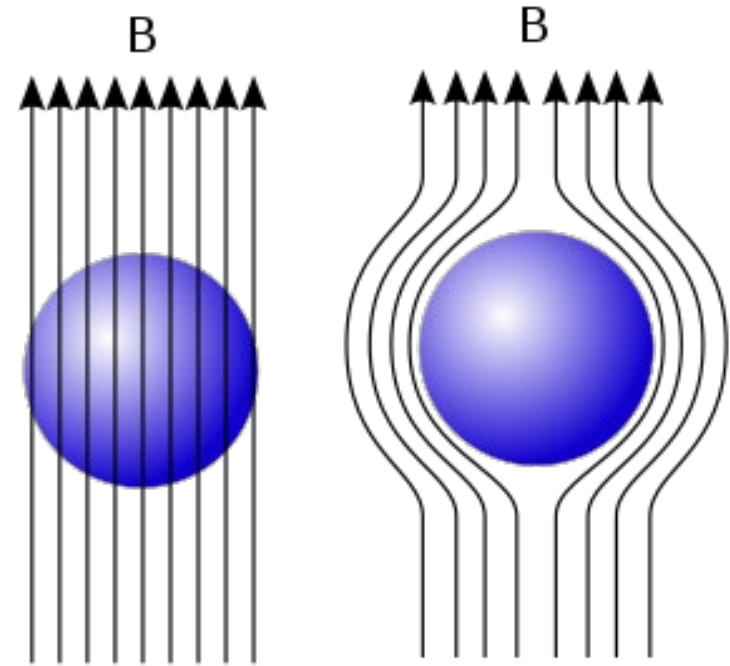
💡 *This is for you — no submission.*



How magnetic field behave in material media?

Magnetic field lines prefer to go through some materials but bypass other materials.

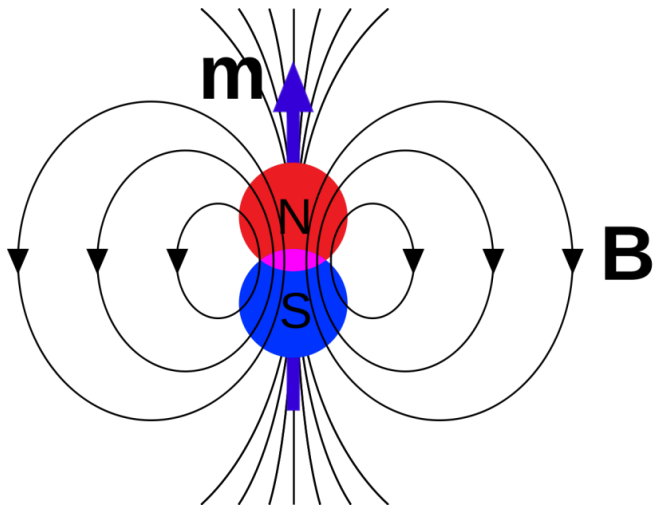
Why so? What causes this behaviour?



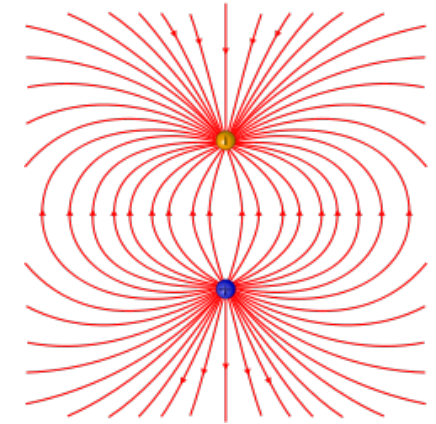
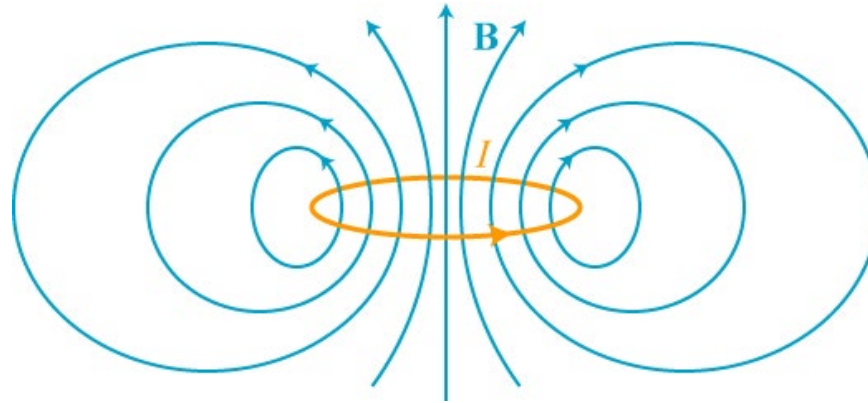
Magnetic dipoles

A *magnetic dipole* is a pair of poles (N and S) separated by a very small distance. It is analogous to an electric dipole .

A magnetic dipole forms in any small current carrying loop.



A magnetic dipole



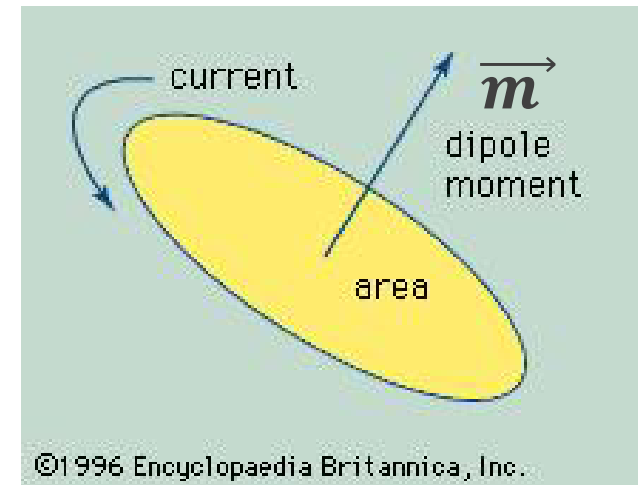
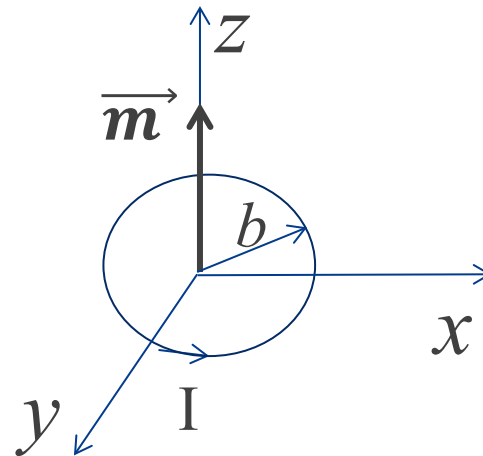
An electric dipole

Dipole moment of magnetic dipole

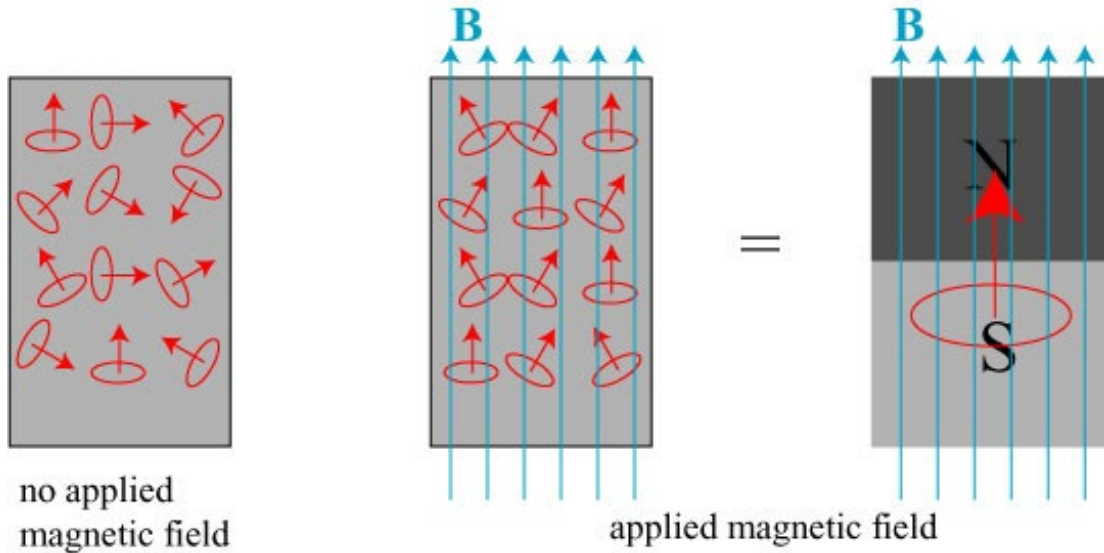
The dipole moment $\vec{m}$ is defined as the product of the current that produces the dipole and the area of the dipole. The dipole moment is perpendicular to the current loop's surface area.

$$\vec{m} = \vec{a}_z I \pi b^2 \quad \text{A-m}^2$$

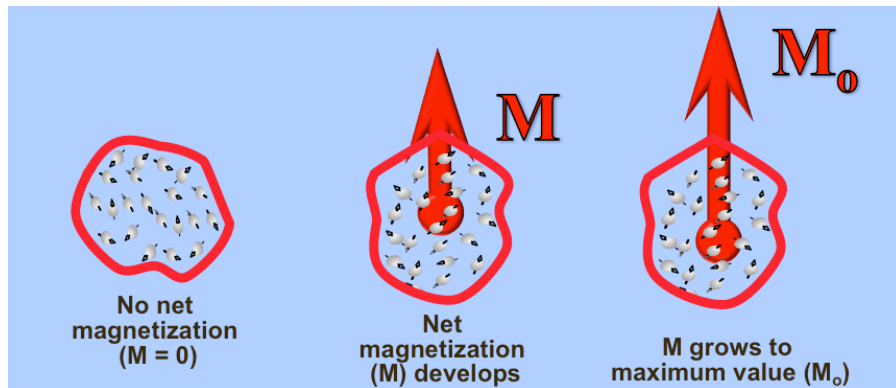
where I is the current in the small loop that forms the dipole and b is the radius of the loop from the centre.



Material media and magnetization vector



- Orbiting electrons of atoms cause circulating currents and form **microscopic magnetic dipoles**.
- The magnetic dipoles in a material have random orientation, resulting in zero net magnetic moment.
- The application of an external magnetic field aligns the magnetic dipole moments.
- Magnetisation: if all dipoles are aligned in the same direction as the applied field, the material is said to have been *magnetised*.



Magnetisation Vector $\mathbf{M}$ is defined as the vector sum of the total magnetic dipole moments per unit volume.

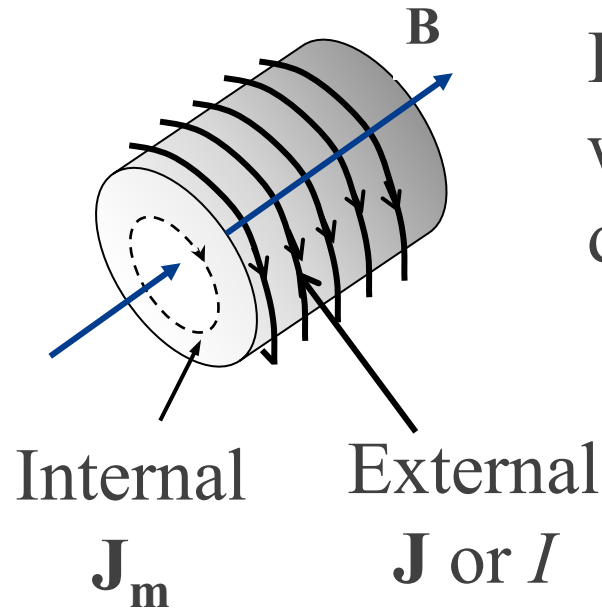
$$\vec{M} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n\Delta v} \vec{m}_k}{\Delta v} \quad (\text{A/m})$$

Magnetization current density $\vec{J}_m$

- The effect of **magnetisation** of a material is equivalent of an induced current density, which is quantified as,

$$\vec{J}_m = \vec{\nabla} \times \vec{M} \quad (\text{A/m}^2)$$

(For full derivation see your textbook section 6-6.)

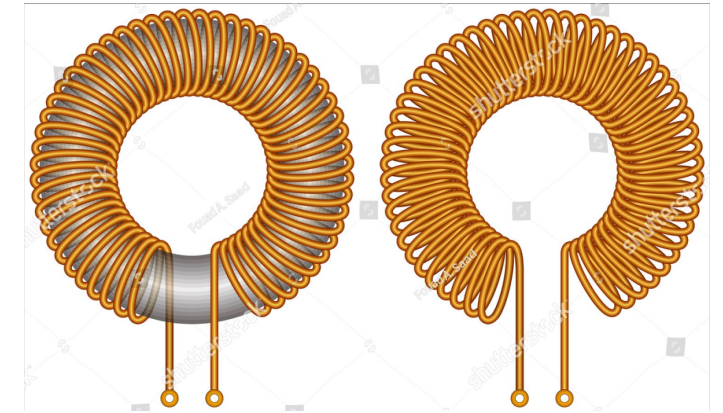


External $\vec{J}$ or I produces the applied magnetic field $\vec{B}$, which magnetises the material and induces internal current density $\vec{J}_m$.

How are $\mathbf{B}$, $\mathbf{H}$ and $\mathbf{M}$ related?

The second postulate includes the induced current density $\mathbf{J}_m$ of the magnetisation when the material media is other than free space ,

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \vec{J}_m \right) \quad \because \vec{J}_m = \vec{\nabla} \times \vec{M} \quad \rightarrow \quad \vec{\nabla} \times \frac{\vec{B}}{\mu_0} = \vec{J} + \vec{\nabla} \times \vec{M}$$
$$\vec{\nabla} \times \left(\frac{\vec{B}}{\mu_0} - \vec{M} \right) = \vec{J}$$



Second postulate in terms of $\mathbf{H}$: $\vec{\nabla} \times \vec{B} = \mu \vec{J} \rightarrow \vec{\nabla} \times \frac{\vec{B}}{\mu} = \vec{J} \rightarrow \vec{\nabla} \times \vec{H} = \vec{J}, \because \vec{B} = \mu \vec{H}$

Comparing two equations of $\mathbf{J}$: $\therefore \vec{H} = \left(\frac{\vec{B}}{\mu_0} - \vec{M} \right)$

Relative permeability of medium

In a linear and isotropic medium $\vec{M} \propto \vec{H} \rightarrow \vec{M} = \chi_m \vec{H}$

where dimensionless proportionality constant χ_m is called **magnetic susceptibility**.

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \rightarrow \vec{B} = \mu_0 (\vec{H} + \vec{M}) = \mu_0 (\vec{H} + \chi_m \vec{H}) = \mu_0 (1 + \chi_m) \vec{H}$$

$(1 + \chi_m) = \mu_r$ is called relative permeability and has no unit.

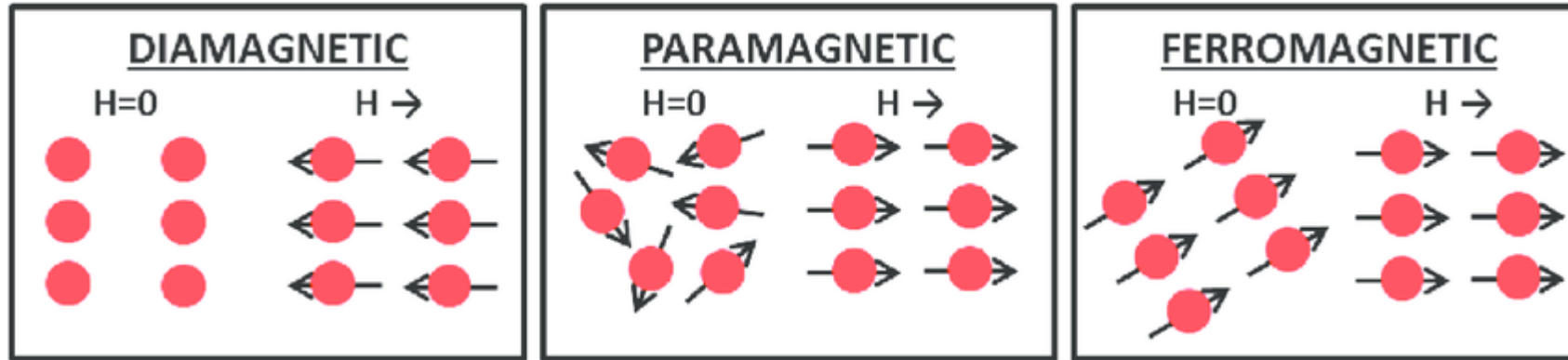
$$\therefore \vec{B} = \mu_0 \mu_r \vec{H} = \mu \vec{H}$$

$\mu = \mu_0 \mu_r$ is called (absolute) permeability and it has the unit of H/m.

Also note Ampere's law in terms of $\mathbf{H}$, $\oint_C \vec{B} \cdot d\vec{l} = \mu I_{enc} \rightarrow \oint_C \vec{H} \cdot d\vec{l} = I_{enc}$

Relative permeability of materials

Some materials are more susceptible to magnetization than others. The value of relative permeability μ_r of a material indicates how susceptible it is to magnetization.



$$\mu_r \leq 1$$

Gold, Silver, Copper, Bismuth etc. Magnetic moment is zero and they are repelled by magnets.

$$\mu_r \geq 1$$

Aluminium, Magnesium, Titanium, air etc. Magnetic moment is not zero but very small.

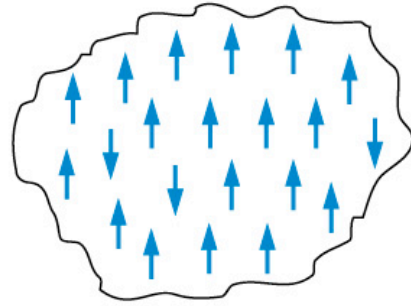
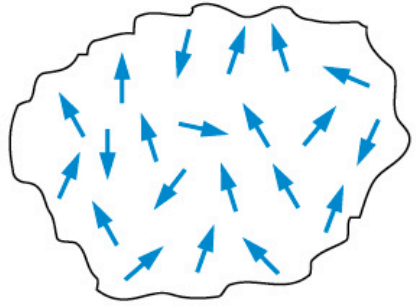
$$\mu_r \gg 1$$

Nickel, Cobalt, Iron, mumetal etc. It has aligned magnetic moments and tends to magnetise easily μ_r could be as large as 30,000 for some alloys.

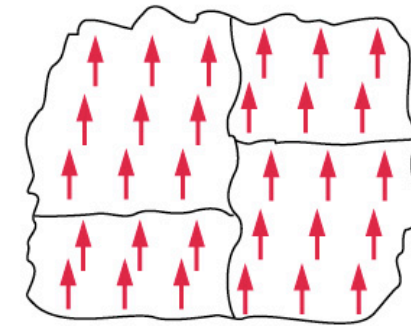
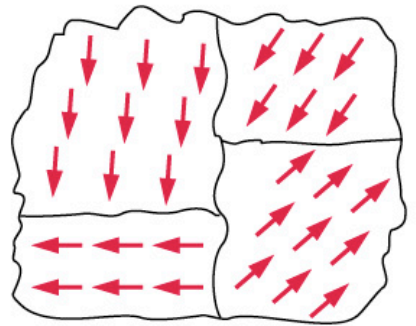
Ferromagnetism- magnetic materials

Magnetic field absent

In presence of magnetic field



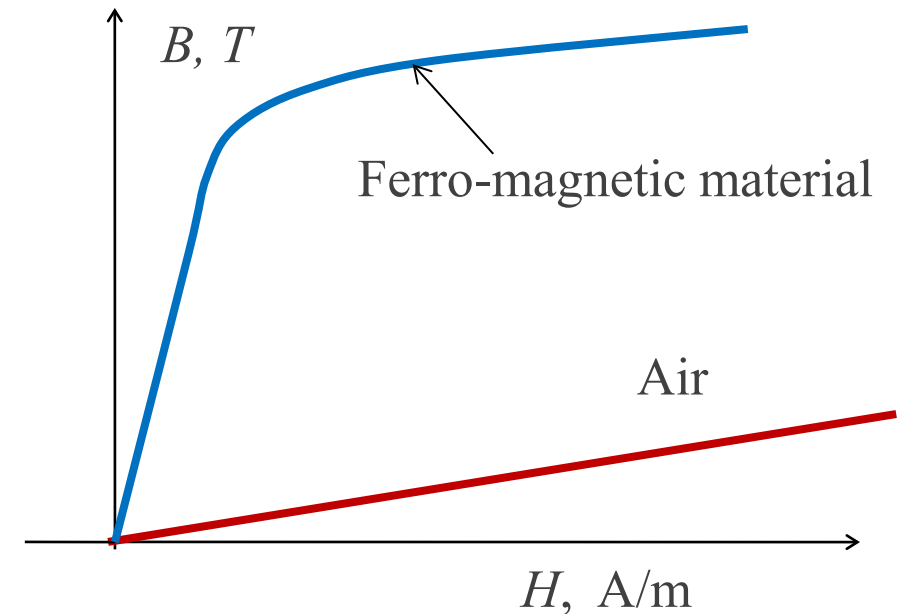
Paramagnetism



Ferromagnetism

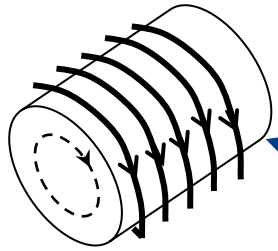
$$\vec{B} = \mu_0 \mu_r \vec{H} = \mu \vec{H}$$

Magnetization (B-H) curve



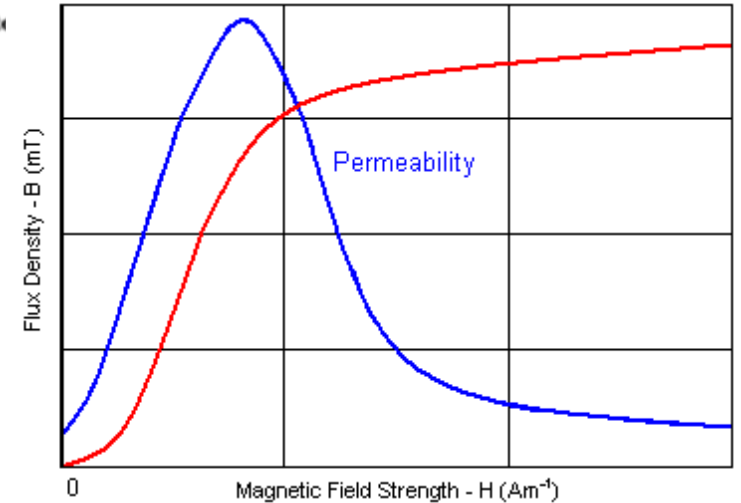
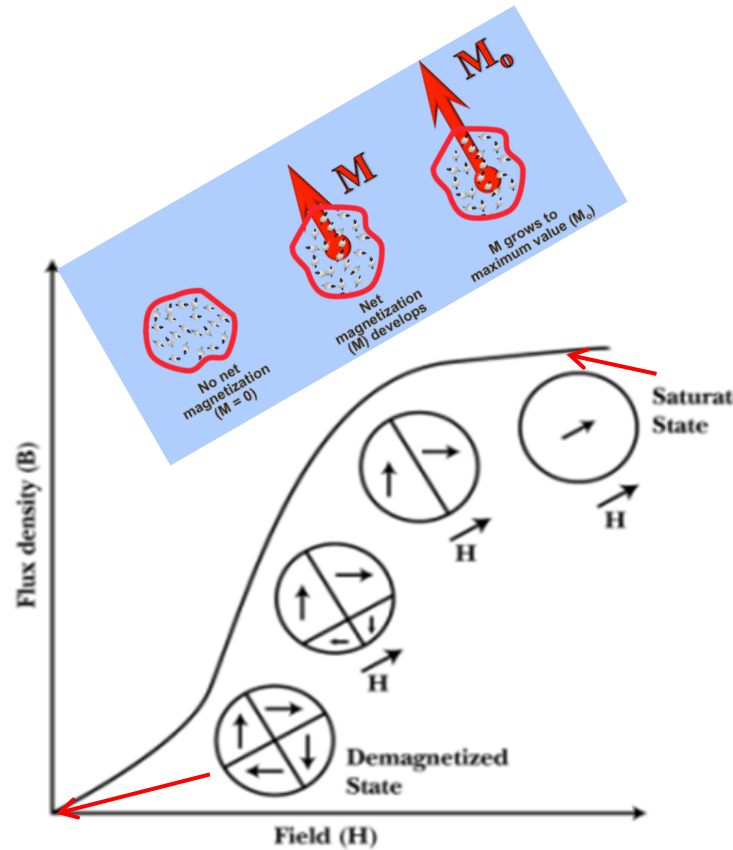
Ferromagnetic materials are composed of domains containing aligned magnetic dipoles. They are also referred to as *magnetic material*. *Ferromagnetic materials* are often used as the core of magnetic devices such as transformer, motors, generator.

Saturation



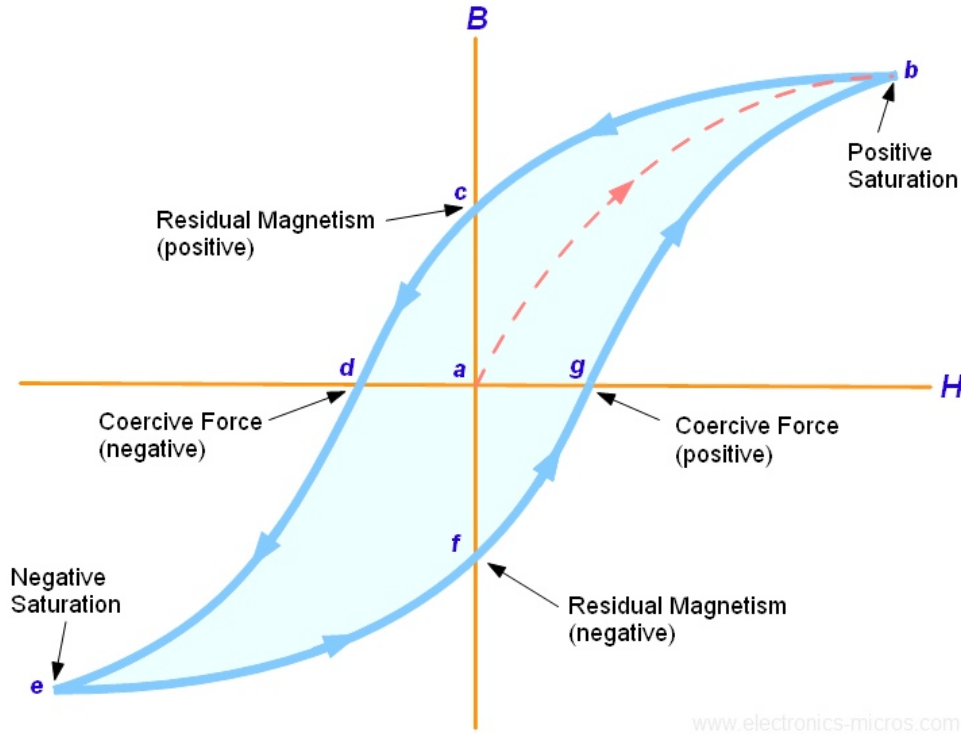
$$\oint_C \vec{H} \cdot d\vec{l} = I_{enc}$$

$$\vec{B} = \mu_0 (\vec{H} + \vec{M}) \rightarrow \vec{B} = \mu_0 \mu_r \vec{H}$$



As the externally applied field $\mathbf{H}$ increases, $\mathbf{M}$ increases and hence, the magnetic flux density $\mathbf{B}$ in the material increases. At some point all the domains are aligned and any increase in the external field does not increase $\mathbf{B}$ any further. This condition is known as **saturated state**. Due to saturation, there is *non-linearity* in the B-H curve of the ferromagnetic material compared to that of air. The slope of the B-H curve at any point gives μ ($= B/H$), but μ is no longer constant when material goes to saturation.

Hysteresis



Hysteresis loop

After magnetisation, when external field H is removed, flux density B does not become zero. This residual flux density is known as remanence (B_r). To make this flux density zero an opposite external field is required, which is known coercivity or coercive force (H_c).

The phenomenon of flux density B lagging behind the applied H is known as hysteresis.

Hysteresis is related to the expansion and collapse of magnetic domains. Hence, it results in loss of energy known as hysteresis loss.

Activity:

Here is a table of properties of some ferromagnetic materials. Which material will you choose for a magnetic device (e.g., a high-power transformer carrying a large current) and why?

Material	Relative permeability μ_r	Coercivity H_c [A/m]	Remanence B_r [T]	Saturation B_s [T]
Iron (0.2% impurity)	9,000	80	0.77	2.15
Pure iron (0.05% impurity)	2×10^5	4	-	2.15
Silicon Iron (3% Si)	55,000	8	0.95	2
Permalloy	10^6	4	0.6	1.08
Superm-alloy	10^7	0.16	0.5	0.79
Permendur	5,000	160	1.4	2.45

Before You Go... (1 minute reflection)

This reflection will count toward optional bonus marks.



0 response submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)



Waiting for response...

Responses will be displayed as a word cloud

Wordcloud

All responses

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1 of 3

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Scan the QR or use link to join



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 Copy link

Key ideas

1 Magnetostatics Foundations

- Two postulates
- Ampere's law
- Calculating B

2 Magnetic Fields in Materials

- Magnetic dipoles
- Magnetisation M
- Define H
- Relative permeability μ_r

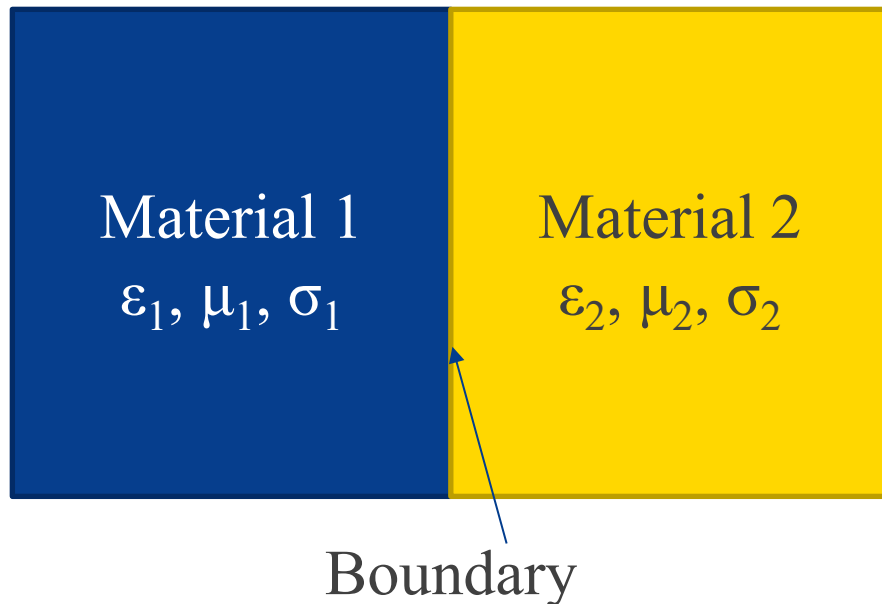
3 Nonlinear Magnetic Behaviour

- B–H curve
- Saturation
- Hysteresis -Energy loss



Q1. How does magnetic field behave at the boundary of two material media?

Q2. Figure below shows two materials with permittivity (ϵ), permeability (μ) and conductivity (σ). List the boundary conditions of the normal and tangential components of the electric field quantity $\mathbf{E}$, $\mathbf{D}$, current density $\mathbf{J}$ and magnetic field quantity $\mathbf{B}$ and $\mathbf{H}$. (You may make a table of two columns – normal comp. and tangential comp. for easy comparison.)



Q3. Comparing the list of Q2, comment on the similarities and dissimilarities in the boundary conditions of Electric and Magnetic fields.

Also, check section 6.10 and examples 6-12 and 3-15 of your textbook by D.K Cheng.

Practice Examples 6-2, 6-7, and 6-8 from the textbook by D.K. Cheng.